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Policy evolution and analysis of characteristics and values of China's marine seed industry development under the food security strategy

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ABSTRACT

Marine seed industry policy is a vital tool for promoting the rapid development of the marine seed industry. Based on 108 policy documents and using social network analysis and co-occurrence analysis, this paper traces the evolution of China's marine seed industry policies since the reform and opening up in 1978. According to policy characteristics, China's marine seed industry policies can be divided into four periods: the period of aquatic resources protection (1978–1998), the period of seedling operation improvement period (1999–2010), the period of modernization and structural adjustment (2011–2016), and the period of comprehensive development driven by science and technology (2017–2022). Policy objectives have changed from being vague and broad to being clear and specific, and the policy effectiveness has shifted from increasing quantity to improving quality and efficiency, which has better promoted the development of the marine seed industry. Based on this, this paper proposes the following strategies for improvement: strengthening scientific and technological innovation within the seed industry, establishing a diversified investment mechanism, creating a fair and open competitive environment, and strengthening the construction of a talent pool in the marine seed industry.

1. Introduction

Seeds are to agriculture what chips are to computers, and the seed industry plays a crucial role in the flourishing of agriculture. Since the 18th Party Congress,¹ the Party Central Committee, led by Comrade Xi Jinping, has made food security a top priority in the governing the country, proposing a new concept of food security to ensure basic self-sufficiency in grains and absolute food security. In 2021 and 2022, the Central Committee issued consecutive calls to "win the battle for seed industry revitalization"² and "fully implement the action plan for seed industry revitalization".³ At the 20th Party Congress,⁴ the Party elevated seed sources to a strategic height related to national security. Currently, the development of the seed industry has received unprecedented attention from the state [1]. In particular, marine seed industry, as the

foundation of the blue granary, is not only essential for national food security and improving people's living standard, but also a crucial way to implement the green development strategy. The possession of marine germplasm resources and the degree of research, development and utilization have also become core indicators of national sustainable development capacity and comprehensive national power [2].

China's aquatic seed industry officially started in the 1950 s, with the artificial propagation of the "four major Chinese carps" [3]. In 1991, the Chinese Aquatic Original and Good Breed Validation Committee was established, initiating the development of the original and good breed system, with a focus on the original and good breed as the mainstay [4]. Since the reform and opening up, with the support of policies at all levels of the central and local governments, China's marine seed industry has undergone a transformation from being nonexistent to thriving.

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¹ The 18th National Congress of the Communist Party of China was held in Beijing on November 8, 2012.

² On February 21, 2021, the Opinions of the State Council of the Central Committee of the Communist Party of China on Comprehensively Promoting the Revitalization of the Countryside and Accelerating the Modernization of Agriculture and Rural Areas, or the Central Document No. 1 of 2021, was distributed.

³ On February 22, 2022, the No. 1 document of the Central Government in 2022, "Opinions of the State Council of the Central Committee of the Communist Party of China on the Key Work of Comprehensively Promoting Rural Revitalization in 2022," was released.

⁴ The 20th National Congress of the Communist Party of China was held in Beijing on October 16, 2022.

Significant progress has been made in both basic research and to germplasm breeding. As seen from the number of marine seed industry policies (Fig. 1), the construction of marine seed industry has received high attention from the country since the beginning of the 21st century, and various policies continue to be developed. In recent years, Guangdong and Hainan have built a "Southern Propagation Silicon Valley", and Shandong has made rapid progress in strengthening the chain in the marine seed industry. The layout of the regional blue seed industry center is beginning to take shape.

Having a clear understanding of the evolution and development of China's marine seed industry policies is essential in establishing a long-term mechanism to support the industry's growth. The objective of this paper is to review the past and present characteristics of marine seed industry policies in China, and identify their value at different stages of development. To achieve this, we will visually analyze policy hotspots to grasp the context of policy development.

Therefore, this paper begins with a brief description of China's marine seed industry evolution and current status in Sections 1 and 2. In Section 3, we introduce the general development of China's marine seed industry policy. Section 4 explains the data source and empirical method used in this paper. In Section 5, we utilize social network analysis, combined with Co-occurrence analysis methods, to provide a detailed analysis of the relevant policies' evolution and describe the characteristics of policy evolution at each stage. Finally, the last section of the paper offers concluding remarks and discussions on our findings.

2. The current development of China's marine seed industry

Thanks to the support of national policies, China's marine seed industry has made significant progress in recent years. This progress is evident in various areas such as the breeding of aquatic species, the protection of germplasm resources, research on breeding technology, and the development of seedling enterprises.

(1) The aquatic seed industry in China is steadily ranked as the world's largest. Currently, China produces 6 trillion aquatic seeds, which have an annual production value of 74.3 billion yuan (approx. US \$10.77 billion). China has also achieved large-scale artificial breeding of more than 300 aquatic economic species [5]. As of July 2022, China has validated 266 new aquatic species, including mirror carp "Longke No. 11". Out of these species, 236 have been bred independently in China, covering major aquaculture species such as fish, shrimp, crab, shellfish and algae [6].

(2) Efforts to ensure the sustainability of marine seedling resources have been significantly strengthened. The Chinese government has been increasing its investment in the construction of aquatic germplasm resource reserves and seed preservation farms. As of now, there are 396 national aquatic germplasm resource reserves and 945 original aquatic breeding farms at or above the provincial level [7]. Special attention has been given to the establishment of genetic breeding centers for marine seedlings, original aquatic breeding farms at all levels, and aquatic breeding bases. Currently, there are 29 existing genetic breeding centers, 87 national original aquatic breeding farms, and 15,000 breeding bases for seedlings [8].

(3) Continuous breakthroughs have been made in research of marine seedling technology. China has continued to make new progress in the technical level of germplasm resource conservation, breeding, and breeding of seed cultivation. It has constructed a large-scale germplasm resource bank, and achieved molecular markers of economic traits for over 30 species of carp, oyster, and kelp [9]. In addition, the National Marine Aquatic Germplasm Resource Bank has established a perfect platform for selection and breeding of good seeds through in-depth collaboration between industry, academia, and research. It has optimized and created a variety of fry breeding techniques, which provides the foundation for the research and development of new aquatic species.

(4) The marine seed industry is expanding in scale. China has seen an increase in the number of registered aquatic seed enterprises to over

19,000. This has resulted in the formation of a modern aquatic seed industry system that combines "conservation, breeding, propagation, promotion, and management" [10]. In 2021, 20 leading enterprises in the integration of breeding, propagation, and promotion of the aquatic seed industry were launched in China [11]. Additionally, in 2022, 121 enterprises were selected by the Ministry of Agriculture and Rural Development as national aquatic seed industry formation-type enterprises, with resources, technology, talent and other elements gathered to key advantageous enterprises [12].

(5) "Marine pasture" has become a new highland for the development of modern seed industry. In 2017, the Chinese Ministry of Agriculture and Rural Development introduced the "Construction Plan for National Marine Pasture Demonstration Area (2017–2025)", aiming to establish 178 national marine pasture demonstration areas by 2025. As of June 2022, 153 such areas have been constructed across the Bohai Sea, Yellow Sea, East Sea and South China Sea, effectively creating a significant opportunity for the development of modern seed industry [13].

(6) The "Blue Medicine Bank" has opened a new chapter in the marine biomedical industry. China's self-developed Alzheimer's disease treatment drug, GV-971, has been approved for marketing, which fills a 17-year gap without new drugs in the field of Alzheimer's disease [14]; Clinical work on a new batch of marine drugs and innovative medical devices, such as BG136, is currently being promoted in an organized manner [15]. Additionally, a variety of products originating from the sea, including special medical food, health food, and pharmaceutical products, are being launched in mass production.

3. Overview of China's marine seed industry policy

We have carefully examined the policy documents pertaining to the marine seed industry that were issued at the national and local levels from 1978 to 2022 (see Fig. 2). The policy texts analyzed in this paper comprise two parts. Firstly, we considered the national policies related to the marine seed industry that were issued by the State Council and national ministries. To obtain this part of the policy, we conducted a search on the websites of the Ministry of Agriculture and Rural Development, as well as other national departments. Secondly, we examined marine seed industry policies that were issued by provincial and municipal governments and departments at all levels. In total, we obtained 123 policy texts, out of which 108 were included in the sample after eliminating those with duplicate content and poor relevance.

Prior to 1980, there was little formal mention of the "marine seed industry" in terms of the number of policies. However, after this time, national and local governments began to issue various systems, including the "Aquatic Seed Management Measures", which clearly aimed to strengthen the selection and breeding of aquatic products and improve the quality of seeds. In 2021, the Seed Industry Revitalization Action Plan was launched,⁵ officially elevating the seed industry to a national strategic level. Geographically, regions such as Shandong, Jiangsu and Zhejiang were early adopters of the marine seed industry, and have since shown industrial advantages in marine seed industry and biological resources development. For instance, Qingdao is focused on the development of a "blue drug library" to help upgrade the city's marine biomedical industry, while Hainan Province's "Southern Propagation Base" offers scientific research, production, sales, scientific and technological exchanges, transformation of results as a comprehensive set of functions. The base aims to breed excellent varieties to ensure national Food security. Similarly, the Rudong Base of Fisheries Research Institute of Jiangsu Province relies on rich marine resources to establish the living resource bank of characteristic Marine fish.

⁵ Action Plan for the Revitalization of the Seed Industry, reviewed and approved by the 20th meeting of the Central Committee for Comprehensively Deepening Reform convened by President Xi Jinping on July 9, 2021.

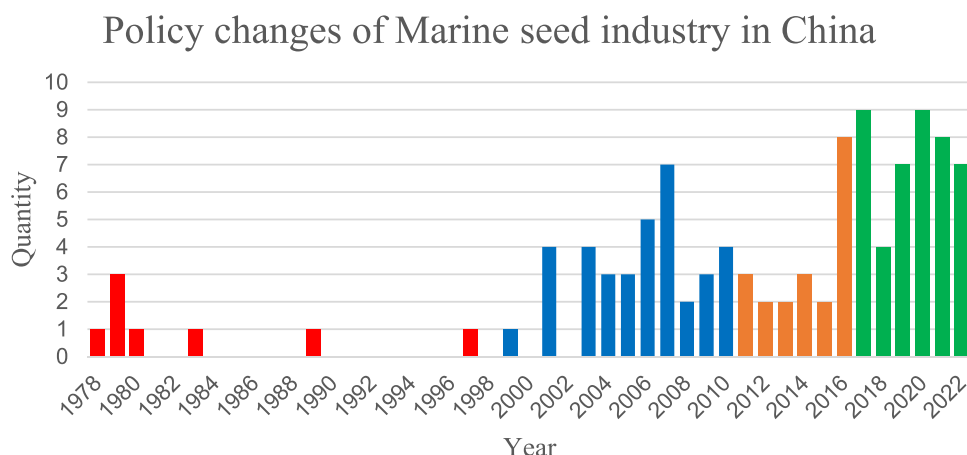


Fig. 1. Change in the number of policies in China's marine seed industry (the four colors represent the four stage of China marine seed industry. Red represents the period of aquatic resources protection (1978–1998), blue represents the period of seedling operation improvement period (1999–2010), orange represents the period of modernization and structural adjustment (2011–2016), and green represents the period of comprehensive development driven by science and technology (2017–2022).

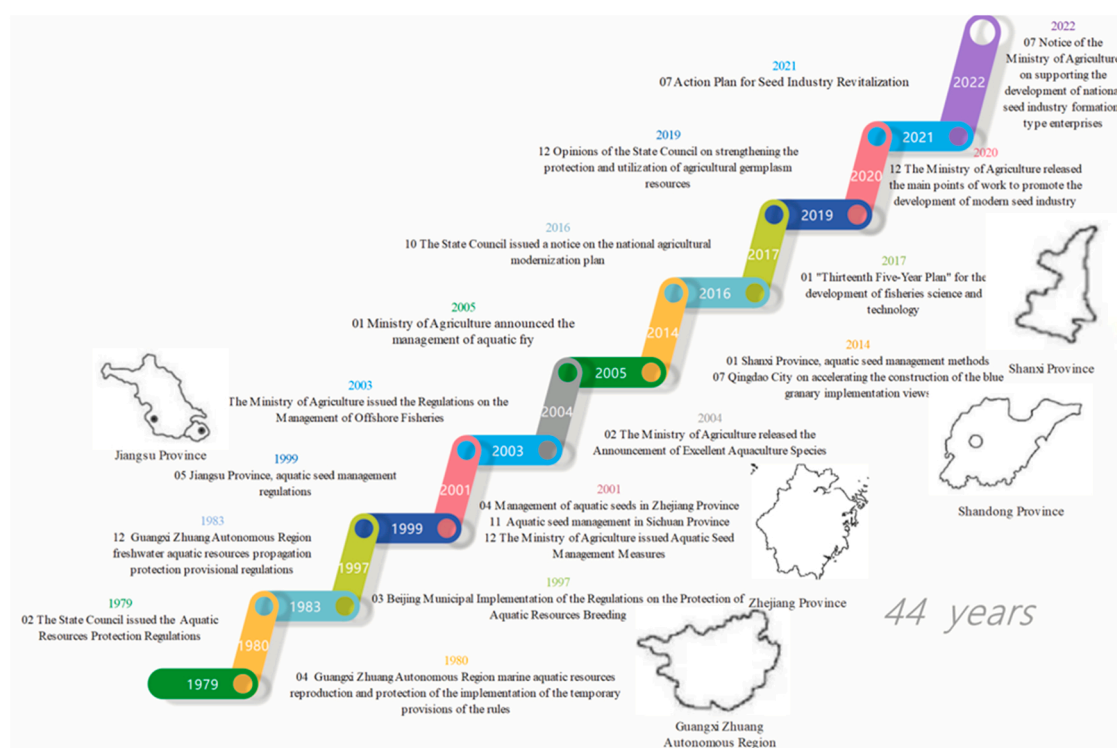


Fig. 2. Timeline of policies related to China's marine seed industry.

4. Data and methods

4.1. Data

We collected 123 national, provincial and city level documents related to "Marine seed industry", "aquatic seed industry", "blue seed industry", "Marine pasture" and "blue seed industry capital".⁶ After systematically screening and sorting out the same documents, 108 documents were obtained as research objects. We used the method of the Jieba algorithm of Python to extract the keywords of each document by year. 4–6 keywords with the highest occurrences were selected each

year, which illustrated the characteristics of the policies of Marine seed industry. Then, we conducted social network analysis method to analyze the relationship of the keywords, and also explained the evolution process of the policies in different periods.

4.2. Methods

This paper aims to visualize and analyze the dynamic evolution process of policy hotspots related to the marine seed industry using the social network approach. By utilizing data on network nodes and connections, the social network analysis method investigates the overall characteristics and interrelationships of the network, and highlights the significance of individual nodes through the calculation of relevant quantitative indicators [16].

The dynamic evolutionary process of marine seed policies involves two main steps. First, policy keywords were extracted using Python's Jieba word separation algorithm, and the number of keywords and

⁶ Data sources contain the websites of Ministry of Agriculture of China (<http://www.moa.gov.cn/>), Gov.cn (<http://www.gov.cn/>), provincial and city governments (e.g. <http://www.jiangsu.gov.cn/>, <http://www.qingdao.gov.cn/> and so on).

policy text content were determined. Second, the co-occurrence of policy keywords was analyzed. By creating a co-occurrence relationship network diagram, the importance and relevance among the policy hotspots were determined based on the node size and connection weights [17]. To visualize the network, Gephi software is selected as a tool as it can address the redundancy of keywords and refine and classify keyword communities to provide references for assessing policy hotspots [18].

The main principle behind Gephi tool is its modularity algorithm, which categorizes nodes based on weights of their connections. This algorithm has been enhanced through the introduction of the Fast-Unfolding Algorithm, which speeds up the computation process. It specifically involves two stages.

In the first stage, assume that there is a weighted network consisting of N nodes. First, a different community is assigned to each node. Then, for each node i and its neighboring node j , the modularity gain is evaluated that results from removing node i from its current community and adding it to j 's community. Node i is then placed in the community that yields the highest gain. The process is repeated sequentially for all nodes until no further improvements can be made. If a node does not yield any gain by moving to another community, it remains in its original community. When node i is moved to community E , the modularity gain ΔQ is calculated using the following expression:

$$\Delta Q = \left[\frac{\sum \text{in} + 2k_{i,\text{in}}}{2m} - \left(\frac{\sum \text{tot} + k_i}{2m} \right)^2 \right] - \left[\frac{\sum \text{in}}{2m} - \left(\frac{\sum \text{tot}}{2m} \right)^2 - \left(\frac{k_i}{2m} \right)^2 \right] \quad (1)$$

In Eq. (1), ΔQ is the modularity gain that results from node i moving into community E ; $\sum \text{in}$ is the sum of the weights of the internal edges of community E , which is the number of keyword co-occurrences. $\sum \text{tot}$ is the sum of the weights of all edges of the nodes contained in community E ; k_i is the sum of the weights of the edges of node i ; $k_{i,\text{in}}$ is the sum of the weights of the edges of node i that are connected to the internal nodes of community E ; and m is the sum of the weights of the edges of the overall network, i.e., the sum of the co-occurrence times of all keywords with each other.

In the second stage, the communities obtained in the first stage are used as new nodes to reconstruct the subgraph. The weight values of the edges between the two new nodes are calculated as the sum of the weight values of the edges between the corresponding two communities. Once the second stage is completed, the first stage of the algorithm can be reapplied to the obtained weighted network and iterated until no further changes occur, and the maximum modularity value is reached [19]. This process yields a modular classification of hot words for marine seed-related policies during the corresponding period.

5. Value analysis of marine seed industry policy features

By conducting a comparative analysis of key words of marine seed industry policies through different periods and examining the timing of landmark policies and events, the evolution of China's marine seed industry policies can be divided into four phases⁷: the period of aquatic resources protection (1978–1998), the period of seedling operation improvement period (1999–2010), the period of modernization and structural adjustment (2011–2016), and the period of comprehensive development driven by science and technology (2017–2022). The dynamic evolution process of each period is reflected by the policy hotspot classification network diagram based on modular results, in which the size of nodes reflects the importance of policy keywords, the color of nodes represents the specific classification results, and the size of the connection between nodes reflects the frequency of keyword co-

occurrence.

5.1. The period of aquatic resources protection (1978–1998)

Prior to 1998, the marine seed industry was not given much attention in policy documents. However, after China's reform and opening up, efforts were made to protect marine aquatic resources by enforcing fishery production order, cracking down on illegal fishing, and implementing other measures. These actions established a legal framework for the subsequent development of the marine seed industry. This period is therefore referred to as the fundamental period for the development of China's marine seed industry, which was marked by a focus on protecting aquatic resources through policies such as water area protection, state management and species protection (see Fig. 3).

The protection of water area includes "conservation", "aquaculture", "water area", "species" and other related keywords. Representative policies include the Regulations on the Reproduction and Protection of Aquatic Resources (The State Council 国务院, 1979), the General Order of The State Council on the Protection of Reservoir Security and Aquatic Resources (The State Council 国务院, 1979), have been implemented to address the issue. Fig. 3 shows that conservation, reproduction and water area are the most frequently co-occurring keywords. The Regulations on the Reproduction and Protection of Aquatic Resources (The State Council 国务院, 1979) specifically aim to protect the parents and juveniles of economically valuable aquatic animals and plants, as well as the aquatic environment in which they reproduce and grow. Local governments have also taken measures to improve water conditions and conduct artificial seeding, depending on local conditions. These measures ensure that the marine seed industry has access to good water resources, which supports the sustainable development in this field.

State management includes "state", "safety", "fishing boat" and other related keywords. One of the representative policies is the Request Report of The State Council to the State Fisheries Administration on Improving the State-run Marine Fishery Management System (The State Council 国务院, 1978). Since 1949, China's marine fishery has developed rapidly, and the modernization of fishing boats and other operating facilities has enabled cross-provincial and cross-sea operations [20]. In this stage, the primary focus of the policy is to reform the state-owned fishing enterprises through implementation of unified leadership and hierarchical management, in order to better adapt to changes in the marine fishery market environment.

Variety protection includes "economy", "value", "variety" and "protection money" and other related keywords. The representative policy is the Circular of the Ministry of Agriculture on the Determination of the

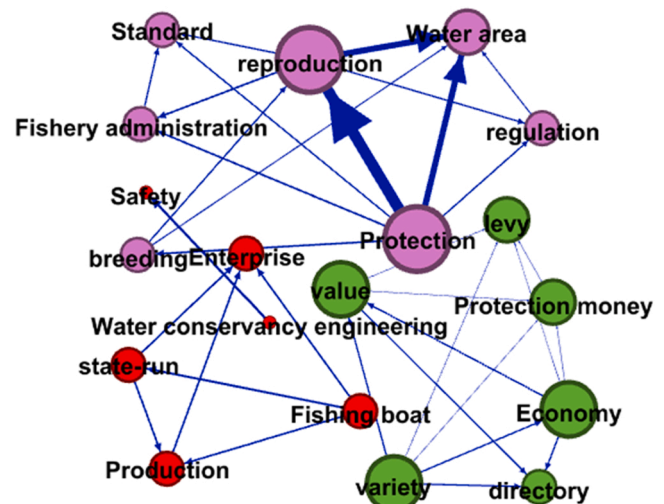


Fig. 3. Aquatic resources protection period policy hotspot network map.

⁷ In Figs. 3–6, different colors represent different policy tendencies in each period to better show the dynamic change process of policy hotspots in each period.

Catalogue of fishery resources with higher economic value (Ministry of Agriculture农业部, 1989). Recognizing the decline of offshore fishery resources and ecological damage caused by excessive fishing, China implemented protection fees on fishery resources with high economic value, such as large yellow croaker, grouper and prawn, in accordance with the Measures on the Collection and Use of Protection Fees for the Multiplication of Fishery Resources (The State Council国务院, 1988). Since 1989, this approach has aimed to enhance the value and protection of fishery resources through economic means.

5.2. The period of seedling operation improvement (1999–2010)

Since 2000, the world has increasingly focused on marine resources, and the scientific development and utilization of these resources have become crucial for China's efforts to enhance its comprehensive national strength [21]. In 2001, the Ministry of Agriculture issued the "Measures for the Management of Aquatic Seedlings", which clearly aimed to strengthen the breeding of aquatic varieties, improve the production and management of aquatic seedlings, and ensure the quality of aquatic seedlings. This demonstrates that the Chinese government places great importance on the safety of aquaculture. We refer to this stage as the seedling management and regulation period of China's marine seed industry. Policy hotspots include the management of aquatic seedling and the promotion of healthy aquaculture practices for aquatic products (see Fig. 4).

Aquatic seedling management, mainly involves keywords such as "seedling", "germplasm", "quality" and others. Representative policies include "Management Measures for Aquatic seedlings" (Ministry of Agriculture 农业部, 2001), "Announcement of Excellent aquaculture Varieties" issued by the Ministry of Agriculture (Ministry of Agriculture 农业部, 2004). Initially, China's aquaculture industry prioritized development over protection" [22], resulting in a low coverage rate of new aquatic products, around 25% [23]. However, to promote the sustainable development of aquaculture industry, the cultivation of aquatic seedlings became the new policy orientation at this stage.

Healthy aquaculture of aquatic products includes "aquaculture", "demonstration area" and other keywords. Representative policies are mainly Regulations on Quality and Safety Management of Aquaculture (Ministry of Agriculture 农业部, 2003), and Implementation Plan for Promoting Healthy Aquaculture in 2008 (Ministry of Agriculture 农业部, 2008), aiming to realize environmentally friendly development of aquaculture [24]. Additionally, the Seed Law (The State Council 国务院, 2000) was promulgated, which introduced systematic and standardized measures for measuring environmental impact, drug delivery standards, production monitoring and other aspects of aquatic breeding.

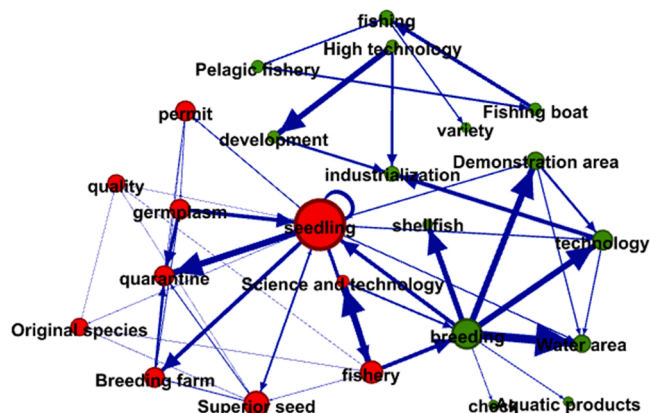


Fig. 4. Policy hotspot network of seedling management and regulation period.

5.3. The period of modernization and structural adjustment (2011–2016)

At the 18th National Congress,⁸ China put forward the strategy of becoming a maritime power for the first time, setting a new tone for sustainable, ecologically sound and environmentally friendly marine fisheries. The 12th Five-Year Plan for National Fishery Development (Ministry of Agriculture 弄业务, 2011), along with other policies, provide clear answers on how to achieve this goal. Specifically, the plan emphasizes the need to strengthen scientific and technological support, ensure fishery capacity, build "five industrial systems" for aquaculture industry, value-added fishery, fishing industry, processing industry and recreational fishery, and promote the modernization of the fishery industry. Therefore, we define this stage as the period of modernization structural adjustment for China's marine seed industry with policy hotspots including marine ranching, modern fishery breeding and modern fishery reform (see Fig. 5).

Marine pasture includes "modern agriculture", "pasture", "demonstration area" and other keywords. One of the representative policies is the Notice of the Ministry of Agriculture on Establishing National Marine Pasture Demonstration Zones (Ministry of Agriculture 农业部, 2015). Marine ranching is a modern approach for fishery production that plays a crucial role in protecting and increasing fishery resources, as well as restoring the ecological environment of waters [25]. To achieve this, artificial reef construction, proliferation and release are used to build marine pastures that integrate aquatic seedling rearing and aquaculture. These efforts help to conserve fishery resources and endangered species, ensure the cultivation of seedlings from the source, and improve the breeding level and quality. For instance, Rizhao City in Shandong Province, has built 7 marine management and protection platforms with a total of 3.895 million cubic meters of reef building, relying on 19 provincial-level marine ranches. These efforts have created suitable conditions for spawning and breeding of fish and shrimp, developed recreational fisheries, driven radiative seedling rearing, processing, sales and other related industries, and promoted the comprehensive development of marine ranches [26].

Modern fishery and aquaculture include "aquatic products", "aquaculture", "modern", "ecological" and other keywords. Representative policies that focus on the development of the fishery industry in China include the 12th Five-Year Development Plan for National Fishery

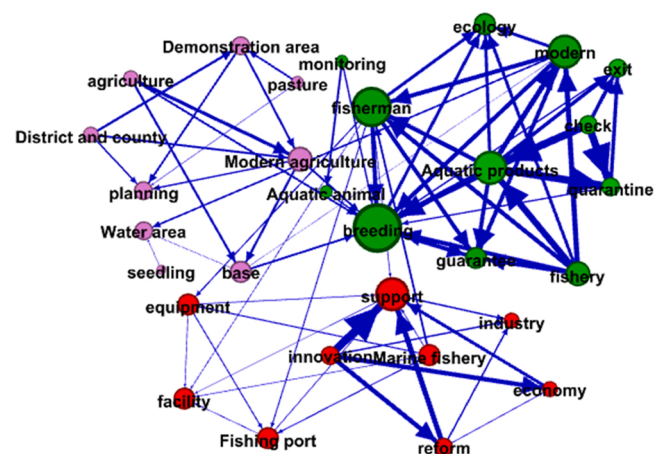


Fig. 5. Policy hotspot network during the modernization Structural adjustment period.

⁸ The 18th National Congress of the Communist Party of China was held in Beijing on November 8, 2012.

(Ministry of Agriculture 农业部, 2011) and the Several Opinions of The State Council on Promoting the Sustainable and Healthy Development of Marine Fishery (State Council 国务院, 2013). These policies present specific ideas for constructing a modern fishery industry, which involves transforming the traditional fishery production methods and leveraging the benefits of digital technology, such as using artificial intelligence to promote intelligent farming [27]. The integration of digital technology has greatly promoted the research and development innovation within the marine seed industry and has laid the groundwork for realizing the potential of the "headquarters economy" of marine seed industry.

Modern fisheries reform includes "Marine fisheries", "reform", "innovation", "support" and other keywords. The representative policies include the Guiding Opinions of the Ministry of Agriculture on Promoting the Transformation of Fishery Mode and Structure (the Ministry of Agriculture 农业部, 2016), the Implementation Plan of the Three-Year Action of "Internet + " Modern Agriculture (the Ministry of Agriculture 农业部, 2016). These policies aim to strengthen the protection of seedling resources through regulatory policies, fiscal means and financial markets, and build aquatic germplasm resources reserves. Various financial innovative products have been launched to expand financing channels and provide financial support for the marine seed industry. For instance, Agricultural Bank of China has actively innovated guarantee products to address the fund bottleneck of core research and development of the seed industry of Guangdong Haida Group [28].

5.4. The period of comprehensive development driven by science and technology (2017–2022)

The interweaving of the century's epidemic and changes has presented significant constraints and challenges for China's marine fishery industry. Enhancing the capacity to produce high-quality aquatic products and addressing the shortcomings of marine breeding technology have become essential priorities for China's fishery development during this period. Therefore, we define this time as the period of technology-driven comprehensive development of China's marine seed industry. Policy hotspots include breeding innovation, ecological

aquaculture and public-private collaborative development (see Fig. 6).

Breeding innovation includes "breeding", "innovation", "enterprise" and other keywords. In 2020, the Central Economic Work Conference designate the seed issue as a separate focus,⁹ Highlighting the crucial role of the marine seed industry in developing a healthy and sustainable aquaculture industry [29]. The "14th Five-Year Plan for National Fishery Development" (Ministry of Agriculture 农业部, 2021) emphasizes the need to strengthen the protection and utilization of aquatic germplasm resources, promote innovation in the seed industry and increase investment in breeding enterprises. To this end, Hainan Province is leveraging its natural location advantages to establish a "Southern Silicon Valley" centered around Yazhou Bay Science and Technology City in Sanya, creating an innovation platform that brings together government, research institutions and enterprises [30].

Ecological breeding includes "pasture", "breeding", "ecology", "demonstration area" and other keywords. Representative policies include the Construction Plan of National Marine Ranch Demonstration Zones (2017–2025) (Ministry of Agriculture 农业部, 2017), and Management Standards of National Marine Ranch Demonstration Zones (Ministry of Agriculture 农业部, 2019), which emphasize the principle of "ecological priority" for the construction of domestic Marine ranch demonstration zones. For instance, the Northern Modern Ecological Aquaculture Demonstration Park in Changyi City, Shandong Province, has successfully established an intensive, large-scale and standardized shrimp seedling breeding base, with breeding enterprises acting as the link and breeding demonstration park as the terminal. This has led to a reduction in water pollution and the establishment of the entire industrial chain of marine pastures [31].

Public-private participant (PPP) development includes key words such as "support", "rural" and "revitalization". Representative policies include "Suggestions of the CPC Central Committee and The State Council on Comprehensively Promoting Rural Revitalization and Accelerating Agricultural and Rural Modernization" (Ministry of Agriculture 农业部, 2021) and "Guidelines for Social Capital Investment in Agriculture and Rural Areas" (Ministry of Agriculture 农业部, 2021). These policies seek to encourage social capital investment in innovative seed industry enterprises and improve the innovation ability of seed breeding. In this stage, government guide fund is the primary means of PPP to support the development of the marine seed industry in this stage. For example, in Guangdong Province, scientific research institutions such as South China Agricultural University, Guangdong Academy of Agricultural Sciences, Guangdong Reclamation Group and others have been integrated into Guangdong Seed Industry Group. To support the development of the Seed Industry Group, Guangdong provide has established the Guangdong Seed Industry Revitalization Guide Fund, which has attracted over 10 billion yuan (approx. US\$1.45 billion) in social capital [32].

6. Conclusions and discussions

6.1. Conclusions

This paper employs social network analysis and co-occurrence analysis methods to visually analyze the dynamic migration process of policy hotspots in China's marine seed industry. By tracking the changes of policy hotspots and landmark policies over time, we have identified four stages in the policy evolution process. The changes of policy dynamic migration are described by integrating the social network map as follows:

(1) Characteristics of policy evolution: Firstly, policy objectives have

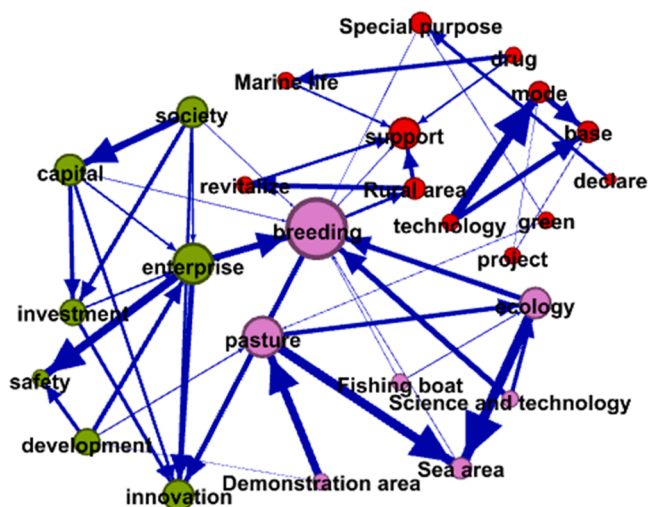


Fig. 6. Policy hotspot network in the technology-driven comprehensive development period.

⁹ 2020 Central Economic Work Conference was held in Beijing from December 16–18, 2020, where Xi Jinping delivered an important speech to summarize the economic work in 2020 and analyze and deploy the economic work in 2021.

shifted from being vague and broad to clear and specific. Since 1999, central and local governments have successively issued "Measures for the Management of Aquatic Seedlings", which has led to the marine seed industry entering a stage of standardized development. Secondly, the policy focus has shifted from quantity to quality and efficiency. During the early stage of the marine seed industry's development, the focus was mainly on achieving higher aquaculture yield. However, in the 21st century, policy makers have become more concerned with the strategic position of the marine seed industry. Priority now includes strengthening breeding capacity and balancing development with environmental protection. Thirdly, policy implementation has moved from a single financial instrument to a more diversified market mechanism. In the past, the government relied mainly on direct subsidies to support the development of the marine seed industry. However, with the acceleration of the marketization process, a commercial breeding system has emerged. This system involves seed industry enterprises, financial institutions, social capital, scientific research institutions and other parties, all operating under the guidance of the government.

(2) Policy effect. Firstly, a system has been established to protect germplasm resources and produce original superior varieties, with reserves and original seed farms at its core. The government actively promotes the construction of aquatic planting resource reserves and seed preservation fields to ensure a suitable living environment for seedlings. Additionally, an aquatic germplasm bank has been built to safeguard the breeding and supply of superior varieties with significant economic and genetic value. Secondly, the capacity for variety improvement and research and development is continuously being enhanced. China's self-cultivated varieties cover the primary aquaculture varieties of both sea and freshwater [33], forming a modern seed industry system that incorporates breeding enterprises as the primary body. This system combines production, education and research, and integrates breeding and propagation. Thirdly, a marine seed industry management system has been established in China. Regulations on the Reproduction and Protection of Aquatic Resources (The State Council 国务院, 1979), Measures for the Management of Aquatic Seedlings (the Ministry of Agriculture 农业部, 2001), and Notice on Supporting the Development of National Seed Formation Enterprises (the Ministry of Agriculture 农业部, 2022) are some of the policy documents that form the fundamental legal framework for managing China's marine seed industry. These policies help to enhance the institutional environment for the domestic marine seed industry's development.

6.2. Discussions

At present, China's marine seed industry has entered a new stage, but there are still some challenges. The depth and breadth of the identification and preservation of excellent germplasm resources are insufficient, for example, there are still 69 kinds of national key protection of economic aquatic plants and animals have not been included [34], and the related achievements of aquatic seed patents are difficult to transform [35]. In 2019, only 7.4% of sales revenue was invested on R&D for China's top 50 seed enterprises [36], which was far below developed countries [37]. Furthermore, supportive policies for industrial development are not in place [38], a stable and sufficient financial support was still not established.

Based on the current achievements and challenges, we have proposed several potential solutions to formulate long-term and practical mechanisms for China's marine seed industry.

(1) Enhancing scientific and technological innovation of seed industry. Accelerating the cultivation of new aquatic species with efficient resource utilization, eco-friendliness, and high quality and diversity through base research and technological innovation in breeding is essential to ensure the safety of aquatic provenance. This will enable larger breeding companies to become the main body of innovation, foster closer collaboration between science and enterprises, and facilitate the establishment of an effective interest linkage mechanism

between seed industry enterprises and scientific research institutions. By creating an industry-university-research integrated breeding system, we can promote the growth of the marine seed industry.

(2) Establishing a diversified investment mechanism for marine seed industry. The focus is on establishing a diverse investment mechanism for the marine seed industry that is "guided by government funds, mainly invested by companies, and supported by social capital". This will enable us to harness the complementary advantages of various production factors and achieve united strength. We plan to establish a subsidy system that promotes superior seed varieties. We will gradually expand the varieties and scope of subsidies, with a particular focus on supporting leading marine seed industry enterprises that are engaged in commercial breeding and applied research and development. Furthermore, we aim to encourage more scientific and technological resources to flow towards large-scale seed industry companies that have a strong presence in the market.

(3) Creating a fair and open environment for competition. Enhance the implementation and supervision of the seedling licensing system to foster a favorable market environment for high-quality seed enterprises. Develop an early warning mechanism for imported seeds and seedlings, rigorously monitoring their quality and taking necessary safety precautions.

(4) Strengthening the construction of personnel in the marine seed industry. The focus is on key areas of the industry and collaborate with scientific research institutes and universities that have a strong R&D capacity to increase the cultivation of skilled professionals in the field of seed and seedling production and enhance the overall quality of workforce. We recognize the importance of providing incentive for talented individuals in order to stimulate their enthusiasm and creativity, which will contribute to the growth of the industry. Therefore, we will implement policies that effectively encourage and reward outstanding researchers. Moreover, we aspire to build an industrial innovation platform that is at par with international standards. We will attract and cultivate exceptional talent from both within the country and abroad. Our ultimate goal is to establish a nationally recognized innovation team with the ability to compete on the global stage.

Data Availability

Data will be made available on request.

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